

Agenda

1. Smallest Number
2. Merge 2 sorted arrays
3. Merge Sort
4. Count Inversion (additional content)

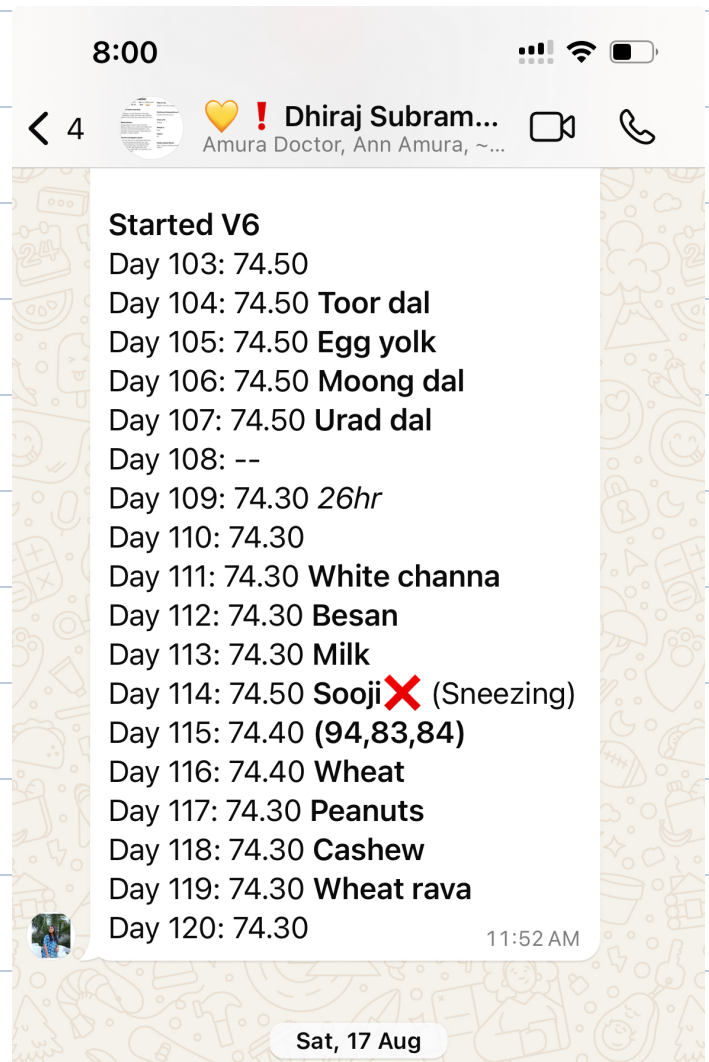
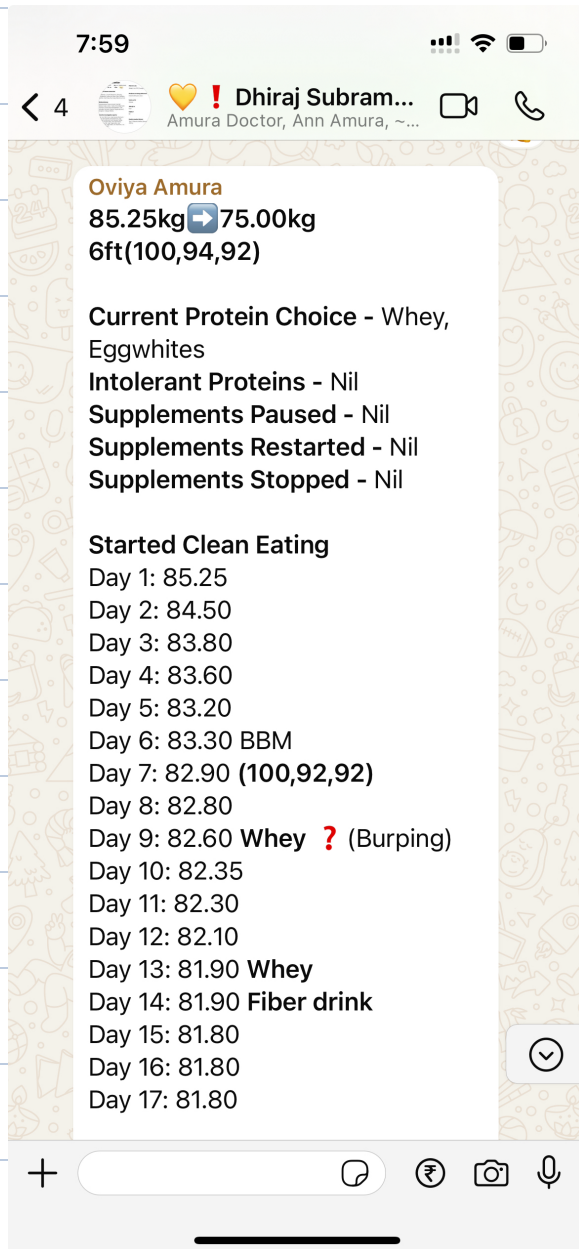
PSP - 69.7 %

Achieve Goals Faster



Elind Kipchoge

Personal story



Group A

(Greater 90)

learner_psp

Sarat Kaza
Aditya Raj
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Ashok Kumar Pradhan
Sai Abhinav Endrapu
Teju Thomas S
Chandrasekar Swaminathan
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Preet Nandeshwar
Rajat Mishra
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Nabil
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Jayesh Jadhav
David Ray
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Sudeep Sharma
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learner_psp

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Group B

(PSP >= 50 , PSP < 90)

learner_psp

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Sambit Mahapatra
Priyanka Kushwaha
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Praveen
Ishan Agarwal
Ashok Bharat Bayana
Naveen Papanna
Abhijit Mandal
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Swapnil Banerkar
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Deepak Kumar
Pravin
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Arghya Biswas
Prem Sai Sameer Namuduru
Syed Md Shakir
Rohit Kumar
Bharath
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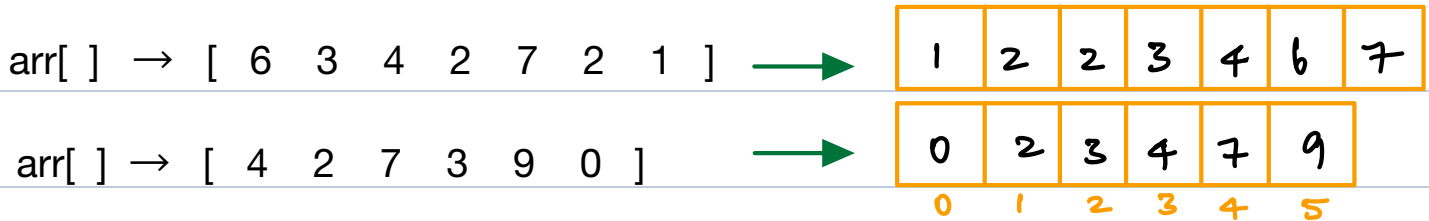
learner_psp

venkat bhargav
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Rishikesh Yadav
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Akhil Balakrishnan
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Anand Krupa Reddy Batthula
Ashish Kumar Verma
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Vivek Asthana



Count Sort

Find the smallest number that can be formed by rearranging the digits given number in an array. Return the smallest number in the form of an array.



Idea -1

sort the array in increasing order

$$T.c = O(n \log n)$$



Idea -2

$A[i] \rightarrow$ stores a digit

smallest = 0

$A[i]$ represents a digit

largest = 9

$$0 \leq A[i] \leq 9$$

Making a frequency array or a hashmap

arr[] → [9 1 2 5 4 2 1 2 5 8]

farr →

0	2	3	0	1	2	0	0	1	1
---	---	---	---	---	---	---	---	---	---

0 1 2 3 4 5 6 7 8 9

This will be my frequency array, my 1 is appearing 2 times, 2 is repeating 3 times, 3 → 0 times, 4 → 1 time and so on

- ① create a array / frequency map of range $[min, max]$
- ② Iterate through the range, and for each freq update the answer in original array



</> Code

```
int Cj smallest No (int Cj arr) {
```

Create a
frequency
array

```
int freq[10] = {0} * 10;
```

```
for (int i=0; i < arr.length; i++) {
```

```
    freq[arr[i]] += 1;
```

// generate sorted array

```
int k=0; Tracking the index of my original array, I'll use K
```

```
for (int i=0; i < 10; i++) {
```

```
    int +-freq = freq[i];
```

```
    for (int j=0; j < +-freq; j++) {
```

```
        arr[k] = i; k++;
```

this for loop will run #of frequency times

```
return arr;
```

T.C = $O(N)$

S.C = $O(\text{range})$

t_freq = freq[i] // for 0th index, freq of 0 is 0 so k remains 0. K only increments if we found a value in the original array

Whenever we found a value we increment k

sum of freq array = n

so k will iterate from 0 to n-1

The above logic is called

count sort

Will count sort work for all inputs? No

We cannot make freq array if $A[i] = 10^9$..so for large values of $A[i]$ we cannot create freq array. Plus this count sort will not work for negative numbers

What is the core idea of Count sort?

①. Create frequency map based on values of array

②. Use freq map to generate ans.

When can we use the count sort?

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



	min	max	Size Hmap	works?
Case I	$A[i] = 0$	$A[i] = 10^9$	4×10^9 4 GB	Not work
Case II	$A[i] = 0$	$A[i] = 10^6$	4×10^6 4 MB	work ✓

if $A[i]$ value is less than or equal to 10^6 --> use count sort



How to handle negative numbers?

arr[] $\rightarrow$ [-2 3 8 3 -2 3]

can you store freq of 11 elements in freq array? Yes

① min = -2, max = 8, range = [-2, 8] $\rightarrow$ 11 elements

② Create f-map of size 11 size 11 means index is from 0 to 10

farr $\rightarrow$

2	0	0	0	0	3	0	0	0	0	1
0	1	2	3	4	5	6	7	8	9	10

Maps

-2	-1	0	1	2	3	4	5	6	7	8
----	----	---	---	---	---	---	---	---	---	---

Here maps is the index to values from -2 to 8

③ How to get index of arr?

$$f_index = arr[i] - (min\ val)$$

$$-2 - (-2) = 0$$

arr -2 -1 0 1 8

f-index 0 1 2 3 . . . 10

④ How to get value from Hmap's index?

$$arr[i] = f_index + (min\ val)$$

with the same formula we can calculate value of arr[i] for hashmap/array



</> Code

```

int CJ sort (int CJ arr) {

```

```

    // get min , max value

```

```

    int minV = min (arr); // inbuilt fn

```

```

    int maxV = max (arr)

```

```

    int range = maxV - minV + 1;

```

```

    int freq [range] = {0} * range;

```

```

    for (int i=0; i < arr.length; i++) {

```

```

        freq [ arr[i] - minV ] += 1;

```

for i = 0, arr[i] = -2, minval = -2; freq[-2-(-2)] = freq[0] will be updated to 1

```

    // generate sorted array

```

```

    int k=0;

```

```

    for (int i=0; i < range; i++) { // for i=0

```

```

        int +-freq = freq [i]; // freq[0] = 2, my 0th index
                                // is repeating 2 times

```

t_freq = 2, this
loop will run 2
times

```

        for (int j=0; j < +-freq; j++) {

```

```

            arr [k] = i + minV; k++;

```

arr[0] = 0 + (-2) = -2

this will repeat for j=1
arr[1] = 0 - (-2)

so both at 0th and 1st index we have -2

```

    return arr;

```

T.C = $O(n * range)$

S.C = $O(range)$

Gmail's All Inboxes Feature

Merge two sorted array

Scenerio

Google's Gmail offers an "All Inboxes" feature that allows users to view emails from multiple email accounts in one seamless interface. This is particularly useful for users managing personal and professional communications through separate accounts.

The feature ensures that emails from all accounts are merged into a single feed sorted by date and time, facilitating better email management and access.

Problem

Develop a function to emulate the "All Inboxes" feature of Gmail.

You are given two sorted arrays that represent timestamps of emails from two different email accounts. Each element in the array is an email object.
Your task is to merge these two arrays into a single list, ensuring that the resulting list is sorted by the timestamp, allowing the user to view emails in a chronological order from both accounts combined.

Example

($-10^9 \leq \text{element} \leq 10^9$)

a[] $\rightarrow$ [2 4 7 8 12] $\rightarrow$ N

Merge two sorted
arrays

b[] $\rightarrow$ [3 5 6 7] $\rightarrow$ M

Idea -1

- ① Create a new array of size $n + m$
- ② Copy each of the elements to the bigger array

arr = [2 4 7 8 12 3 5 6 7]

TC: $(n+m) \log(n+m)$

- ③ Sort the array

arr = [2 3 4 5 6 7 7 8 12]



Sorting - 1

Compare the 1st element of both the arrays. Now $2 < 3$, if $3 < 2$ that means all elements after 3 which are 5, 6, 7 will also be greater than 2. So I don't have to think about checking the rest of the elements in $b[]$ array since the array is sorted.

Idea - 2

Logic

if $a[i] < b[j]$, $2 < 3$
so $arr[k] = A[i]$

$a[] \rightarrow [2 \ 4 \ 7 \ 8 \ 12] \rightarrow N$



$b[] \rightarrow [3 \ 5 \ 6 \ 7] \rightarrow M$



$arr = [2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 7 \ 8 \ 12]$

Similarly, next we compare 3 with 4, if $4 > 3$ then all others elements in $a[] \rightarrow 7, 8, 12$ will also be greater than 3 so I don't have to compare with any elements after 4 in $a[]$. So 3 will be the next element in my $arr[]$



if ($a[i] \leq b[j]$) {

$arr[k] = a[i]$

$i++$; $k++$;

} else {

$arr[k] = b[j]$

$j++$; $k++$;

}

① copy remaining values

</> Code

int[] mergeSortedArray (int[] a, int[] b) {

int i = 0; j = 0; k = 0;

int n = a.length; m = b.length;

int arr[n+m];

while (i < n && j < m) {

if ($a[i] \leq b[j]$) {

$arr[k] = a[i]$; $i++$; $k++$;

}

else {

$arr[k] = b[j]$; $j++$; $k++$;

}

}

// copy remaining value

```
while (i < n) {
```

```
    arr[k] = a[i]; i++; k++;
}
```

```
while (j < m) {
```

```
    arr[k] = b[j]; j++; k++;
}
```

```
return arr;
```

$T.C = O(n+m)$

$S.C = O(1)$

SC is $O(1)$ coz my code for sorting doesnot require any extra space. The resultant array requires space which is asked by the interviewer, this is something not coz of my code

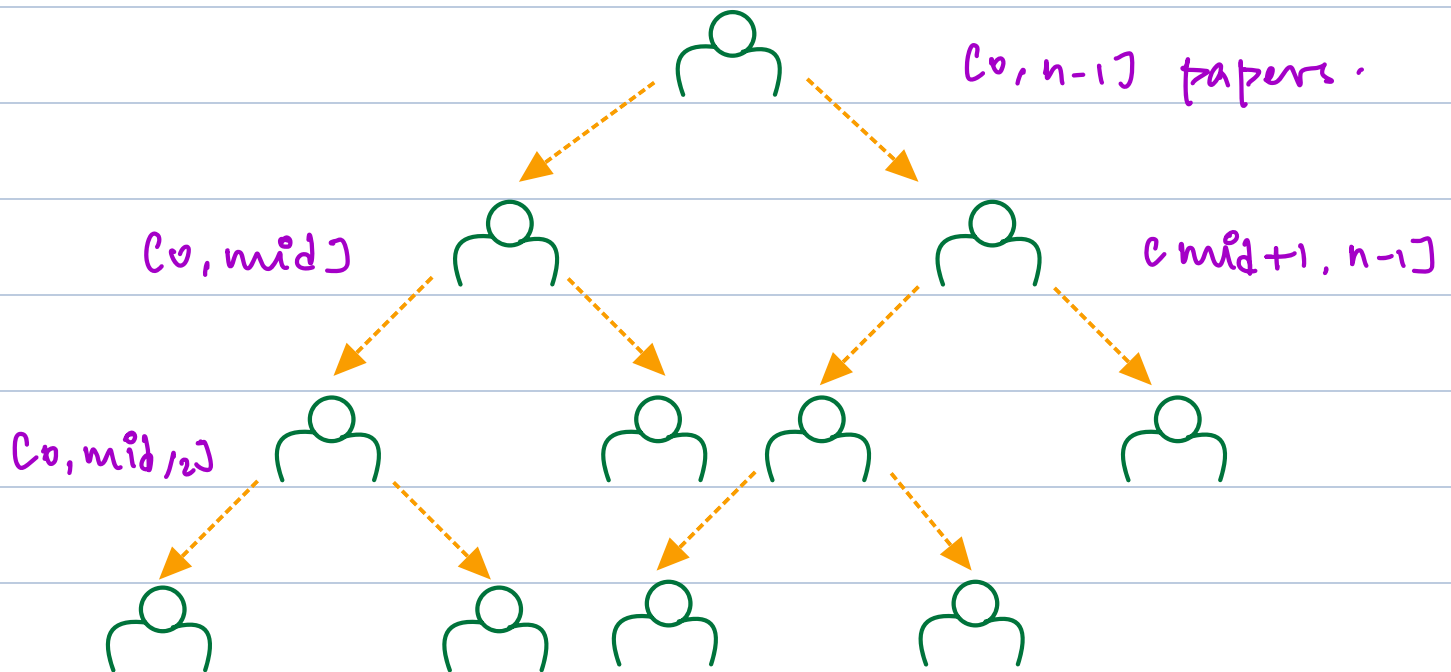
Note :

- ① We now know how to merge two sorted arrays
- ② Can we somehow use this logic to sort the entire array ?

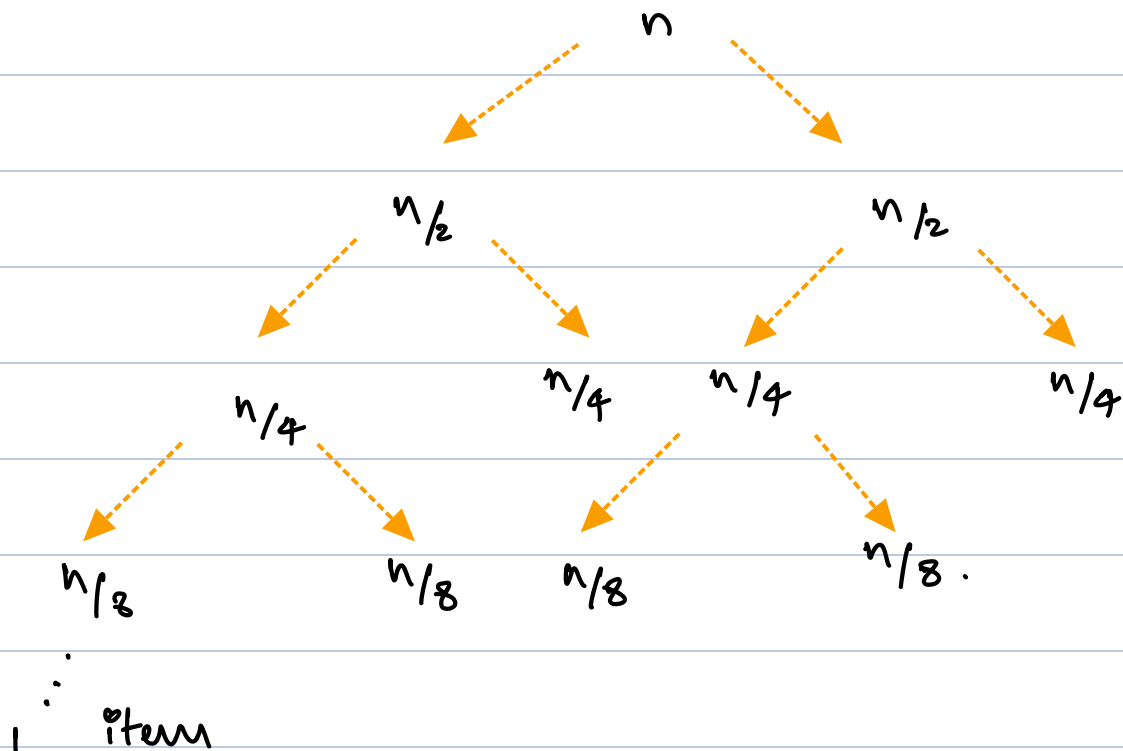
Break : 10:20 pm → 10:27 pm



Merge Sort



Divide and Conquer





3 magic steps

assumptions: `merge (A, l, r)`

main logic: `merge (A, l, mid)`
`merge (A, mid+1, r)` } merge 2 sorted arrays

base logic:

`if (l == r) return`

</> Code

```
void mergeSort (int A[], int l, int r) {
```

```
    if (l == r) return // base condition
```

```
    int mid = (l + r) / 2; // int division
```

```
    mergeSort (A, l, mid);
```

```
    mergeSort (A, mid+1, r);
```

```
    merge (A, l, mid, r);
```

```
}
```

```
void merge (int A[], int l, int m, int r) {
```

```
    int n = r - l + 1;
```

```
    int temp[n] = {0} * n
```

```
    // L array [l, mid]    R array [mid+1, r]
```

```
    int i = l, j = mid+1, k = 0;
```

```
    // run same logic for merging
```



```
while (l <= mid && j <= r) {
```

```
    if (a[i] <= a[j]) {
```

```
        | temp[k] = a[i]; i++; k++;
```

```
    }
```

```
    else {
```

```
        | temp[k] = a[j]; j++; k++;
```

```
    }
```

```
}
```

// copy remaining value

```
while (i <= mid) {
```

```
    | temp[k] = a[i]; i++; k++;
```

```
}
```

```
while (j <= r) {
```

```
    | temp[k] = a[j]; j++; k++;
```

```
}
```

// copy values from temp[] to A[]

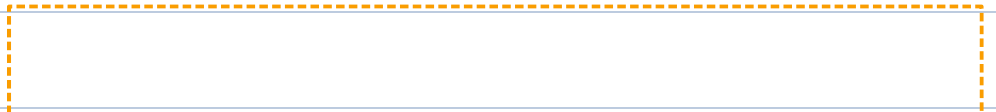
```
k = 0;
```

```
for (i = l; i <= r; i++) {
```

```
    | A[i] = temp[k]; k++;
```

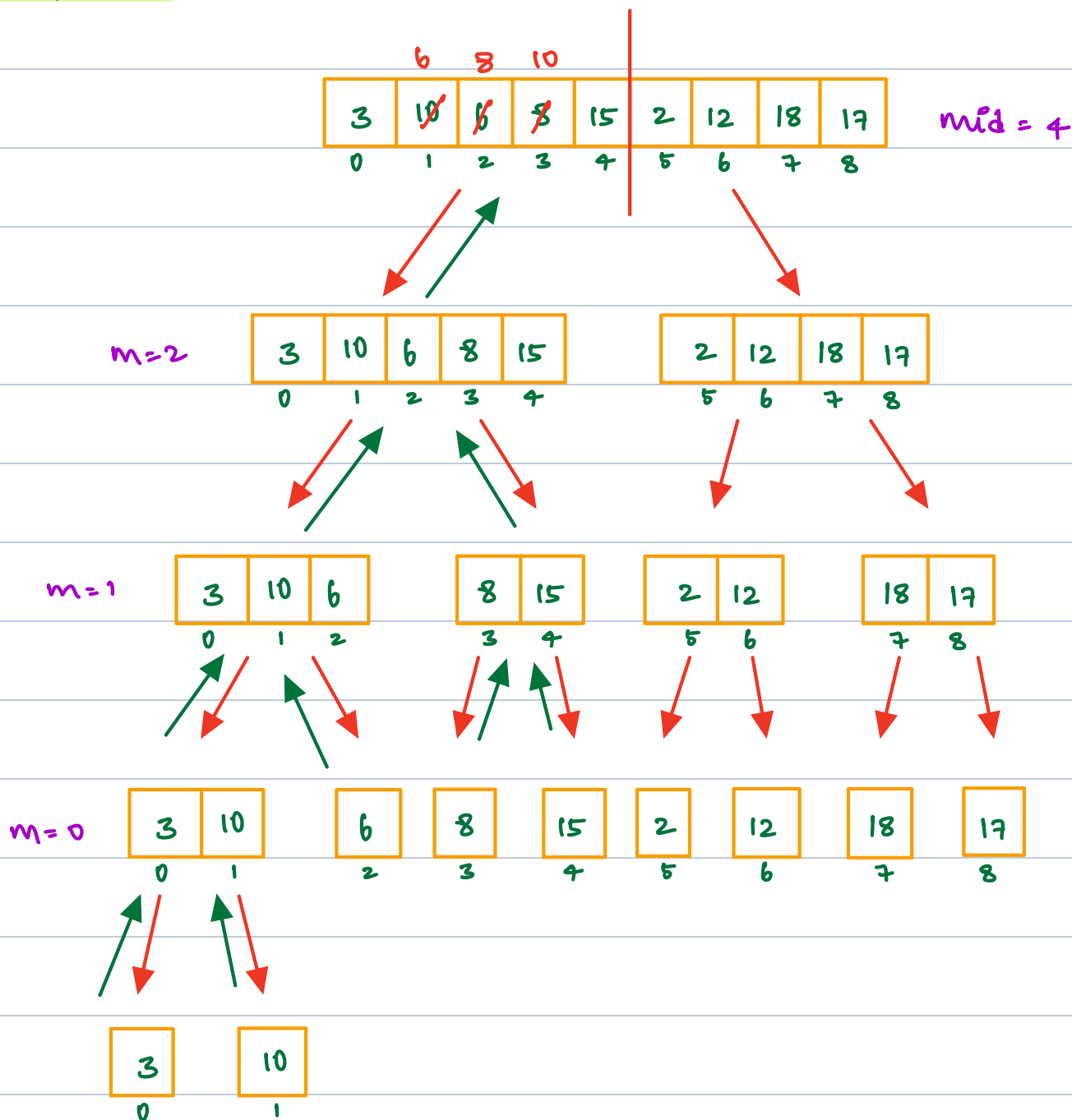
```
}
```

```
}
```



Dry Run

merge sort (A, 0, 8)



Time Complexity Analysis [Merge Sort]

Total number of levels = $O(\log n)$

Total instruction in one level = $O(n)$

T.C = $O(n \log n)$

S.C = $O(n)$

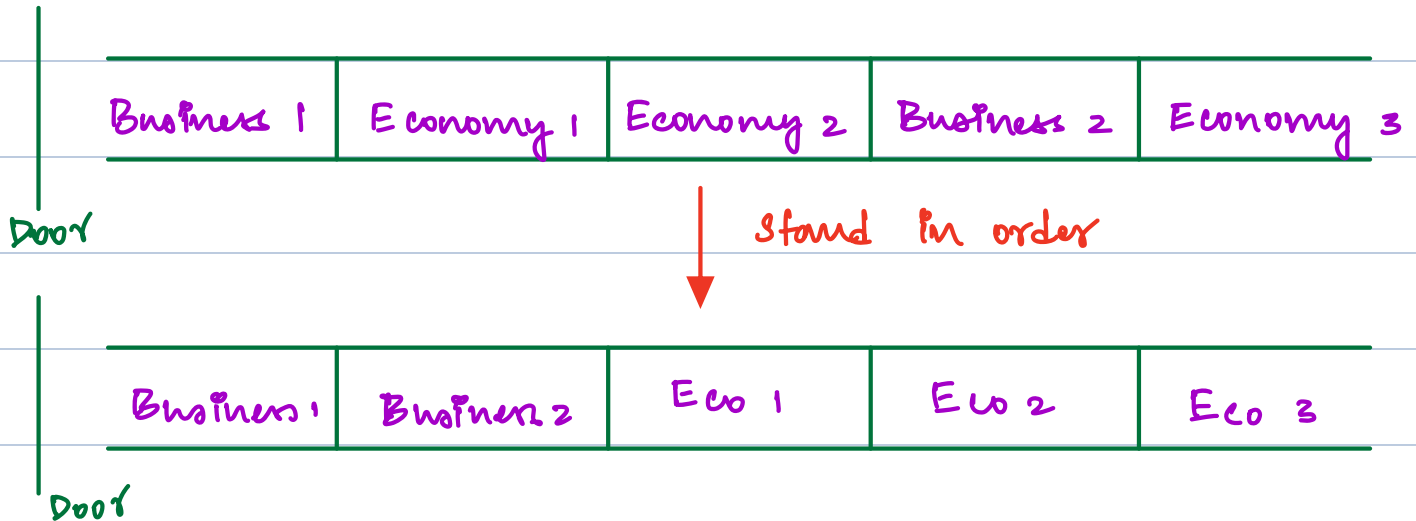
Stable Sort



Does the number with same magnitude appear in the order when it is sorted

Scenario

Airport Line



Count Sort

Not stable

Merge Sort

Stable

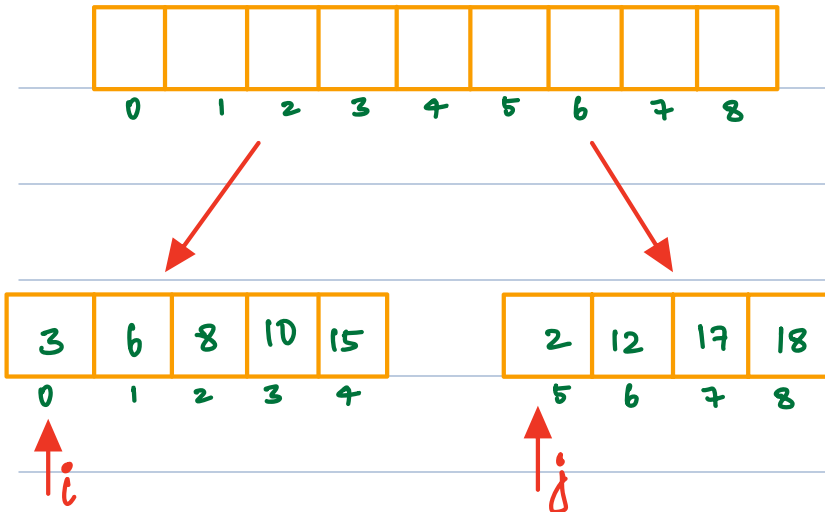
Inplace Sorting

No extra space → $O(1)$

Merge Sort X

Inversion Count

Given an array of integers A. If $i < j$ and $A[i] > A[j]$, then the pair (i, j) is called an inversion of A. Find the total number of inversions of A modulo $(10^9 + 7)$.



if ($a[i] \leq a[j]$)

// not Inversion

else // inversion

$ans += (mid - i + 1)$